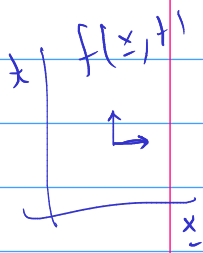


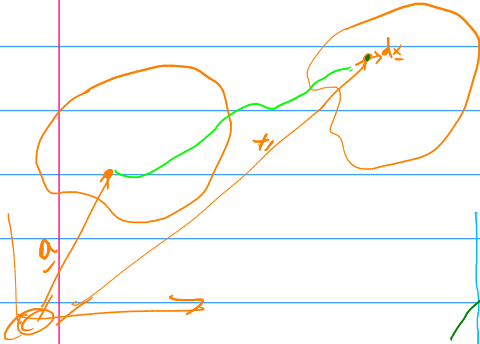
$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad f(\underline{x}) = x_1^2 + x_2 \cdot x_3$$



$$f(\underline{x}, t) = f(\underline{x}(\underline{a}, t), t)$$

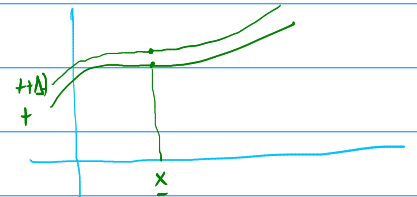
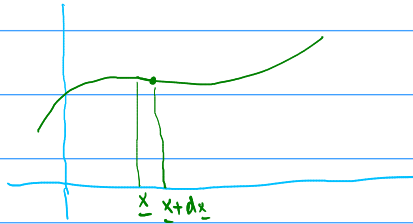
label de la partícula
→ es constante en el tiempo

$$\frac{D}{Dt} f(\underline{x}(t), t) = \left. \frac{\partial f(\underline{x}, t)}{\partial \underline{x}} \right|_{t=fijo} \cdot \frac{\partial \underline{x}}{\partial t} + \left. \frac{\partial f(\underline{x}, t)}{\partial t} \right|_{\underline{x}=fijo} \cdot \frac{\partial t}{\partial t}$$



$$= \left. \frac{\partial f}{\partial \underline{x}} \right|_{t=fijo} \cdot \underline{v} + \left. \frac{\partial f}{\partial t} \right|_{\underline{x}=fijo}$$

convectivo local



Derivada material de un volumen de un conjunto de partículas

$$\frac{D}{Dt} \int_V A dV = \int_V \frac{\partial A}{\partial t} dV + \int_S A v_i n_i dS$$

$$\begin{aligned} \frac{D}{Dt} \int_V A dV &= \int_V \frac{\partial A}{\partial t} dV + \int_V \frac{\partial}{\partial x_j} (A v_j) dV \\ &= \int_V \left(\frac{\partial A}{\partial t} + v_j \frac{\partial A}{\partial x_j} + A \frac{\partial v_j}{\partial x_j} \right) dV \\ &= \int_V \left(\frac{DA}{Dt} + A \frac{\partial v_j}{\partial x_j} \right) dV \end{aligned}$$

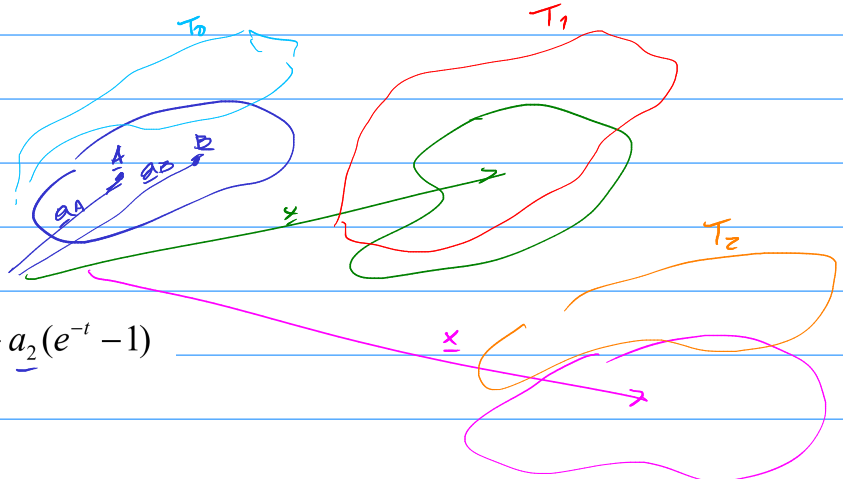
Conservación de masa o Ecuación de continuidad

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho v_j}{\partial x_j} = 0$$

$$\frac{D\rho}{Dt} + \rho \frac{\partial v_j}{\partial x_j} = 0$$

Ejercicio 2:

me define
qué se partícula
se refiere



$$x_1 = \underline{a}_1 e^{-t}, \quad x_2 = \underline{a}_2 e^t, \quad x_3 = \underline{a}_3 + \underline{a}_2 (e^{-t} - 1)$$

$$T(x_1, x_2, x_3, t) = e^{-t} (x_1 - 2x_2 + 3x_3)$$

$$\underline{x} = \underline{x}(\underline{a}, t) \rightarrow T^*(\underline{a}, t) \rightarrow \frac{DT^*}{Dt}(\underline{a}, t) \rightarrow \frac{dT^*}{dt}$$

→ trabajo

$$\frac{dT^*}{dt}$$

$$\frac{DT}{DT} = \nabla_x T \cdot \underline{v} + \left. \frac{\partial T}{\partial t} \right|_{x=cte}$$

Remplacer $a_1 = a_1(x_1, x_2, x_3, t)$

$$\underline{v} = \frac{Dx}{Dt} = \begin{bmatrix} \frac{Dx_1}{Dt} \\ \frac{Dx_2}{Dt} \\ \frac{Dx_3}{Dt} \end{bmatrix} = \begin{bmatrix} -a_1 e^{-t} \\ a_2 e^t \\ -a_2 e^{-t} \end{bmatrix} = \begin{bmatrix} v_1(x) \\ v_2(x) \\ v_3(x) \end{bmatrix}$$

$$\frac{\partial T}{\partial x_1} = e^{-t}$$

$$\left. \frac{\partial T}{\partial t} \right|_{x=cte} = -e^{-t}(x_1 - 2x_2 + 3x_3)$$

$$\frac{\partial T}{\partial x_2} = -2e^{-t}$$

$$\frac{\partial T}{\partial x_3} = 3e^{-t}$$

$f(x, t)$

$$f(x, t) = \frac{DT}{Dt} = v_1 e^{-t} - 2 \cdot v_2 e^{-t} + 3 v_3 \cdot e^{-t} - e^{-t}(x_1 - 2x_2 + 3x_3)$$

Derivée temporelle explicite et terme local

$$\left. \frac{\partial T}{\partial t} \right|_{x=cte} = -e^{-t}(x_1 - 2x_2 + 3x_3) + e^{-t}(v_1 - 2v_2 + 3v_3)$$

$$x_1 = a_1 e^t \Rightarrow a_1 = \frac{x_1}{e^{-t}} = e^t x_1$$

$$x_2 = a_2 e^t \Rightarrow a_2 = x_2 \cdot e^{-t}$$

$$x_3 = a_3 + a_2(e^{-t} - 1) \Rightarrow a_3 = x_3 - x_2 \cdot e^{-t}(e^{-t} - 1)$$

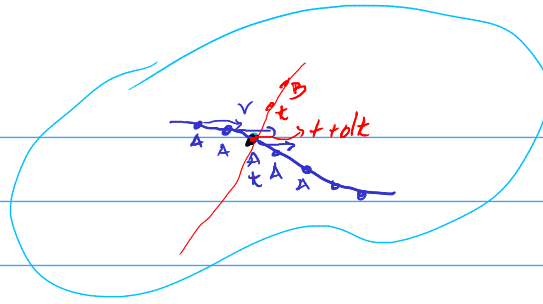
$$\underline{v} = \begin{bmatrix} -e^t x_1 \cdot e^{-t} \\ x_2 \cdot e^t \cdot e^t \\ -x_2 \cdot e^{-t} \cdot e^{-t} \end{bmatrix} = \begin{bmatrix} -x_1 \\ x_2 \\ -x_2 \cdot e^{-2t} \end{bmatrix}$$

$$f(x, t) = \frac{DT(x, t)}{Dt} = -x_1 e^{-t} - 2 \cdot x_2 e^{-t} - 3 x_2 \cdot e^{-3t} - e^{-t}(x_1 - 2x_2 + 3x_3)$$

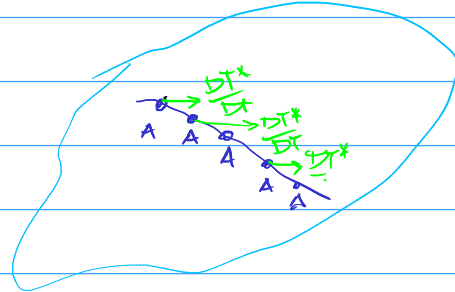
$$= e^{-t} \left(-x_1 - x_1 - 2x_2 + 2x_2 - 3x_3 - 3x_2 \cdot e^{-2t} \right)$$

$$= e^{-t} \left(-2x_1 - 3x_3 - 3x_2 e^{-2t} \right) = -e^{-t} \left(2x_1 + 3x_3 + 3x_2 e^{-2t} \right)$$

$$\frac{DT(x,t)}{Dt} \quad x \rightarrow \text{fijo}$$



$$\frac{DT^*(a,t)}{Dt} \quad a \text{ fijo} \rightarrow$$



$$T(x,t) = e^{-t} (x_1 - 2x_2 + 3x_3)$$

$$x_1 = a_1 e^{-t}, \quad x_2 = a_2 e^t, \quad x_3 = a_3 + a_2(e^{-t} - 1)$$

$$\begin{aligned} T^*(a,t) &= e^{-t} (a_1 e^{-t} - 2a_2 e^t + 3a_3 + 3a_2 e^{-t} - 3a_2) \\ &= e^{-2t} (a_1 - 2a_2 e^{2t} + 3a_3 e^t + 3a_2 - 3a_2 e^t) \end{aligned}$$

$$\begin{aligned} \frac{DT^*(a,t)}{Dt} &= -2e^{-2t} (a_1 - 2a_2 e^{2t} + 3a_3 e^t + 3a_2 - 3a_2 e^t) + e^{-2t} (-4a_2 e^{2t} + \\ &\quad + 3a_3 e^t - 3a_2 e^t) = \\ &= e^{-2t} (-2a_1 + 4a_2 e^{2t} - 6a_3 e^t - 6a_2 + 6a_2 e^t - 4a_2 e^{2t} + 3a_3 e^t - 3a_2 e^t) \\ &= e^{-2t} (-2a_1 - 6a_2 + 3a_2 e^t - 3a_3 e^t) \end{aligned}$$

Si reemplaza en $a = a(x,t)$ obtenida previamente para verificar resultados:

$$\begin{aligned} \frac{DT(x,t)}{Dt} &= e^{-2t} [-2e^t x_1 - 6x_2 e^{-t} + 3x_2 e^{-t} - 3x_3 e^t + 3x_2 e^{-t} (e^{-t} - 1)] \\ &= e^{-t} (-2x_1 - 6x_2 e^{-2t} + 3x_2 e^{-t} - 3x_3 + 3x_2 e^{-2t} - 3x_2 e^{-t}) \\ &= -e^{-t} (2x_1 + 3x_3 + 3x_2 e^{-2t}) \end{aligned}$$